1. The Kohler–Jobin inequality and the transfer to Faber–Krahn

This section is logically self-contained. In Part 1 we reconstruct from scratch, by the Kohler–Jobin rearrangement technique, the spectral shape inequality

$$T(\Omega)\,\lambda_1(\Omega)^{(n+2)/2} \;\geq\; T(B)\,\lambda_1(B)^{(n+2)/2} \qquad (B \text{ any ball}),$$

with equality only for balls. In Part 2 we feed this inequality, together with the global sharp Saint–Venant stability estimate proved elsewhere in the manuscript, into a short and fully explicit computation that yields sharp quantitative Faber–Krahn stability, $\delta_{\mathrm{FK}}(\Omega) \geq c_{\mathrm{FK}}(n)\,\mathcal{A}(\Omega)^2$.

The technique, due to Kohler–Jobin [3, 4, 5], is a spherical rearrangement that, unlike Schwarz symmetrisation, preserves not the *volume* of the superlevel sets but a *torsional* functional of them. We follow the modern exposition of Brasco [1, §§3–5, Thm. 1.1], specialised throughout to the linear case (the Laplacian, $p = 2$, and the Dirichlet energy quotient).

1.1. Notation and scaling conventions.

1.1. Notation and scaling conventions. Throughout, $n \geq 2$ and $\Omega \subset \mathbb{R}^n$ is open with finite measure. We use:

- $u_\Omega \in H_0^1(\Omega)$, the *torsion function*, the unique weak solution of $-\Delta u_\Omega = 1$ in Ω, $u_\Omega = 0$ on $\partial\Omega$;
- $T(\Omega) = \int_\Omega u_\Omega\,dx$, the *torsional rigidity*, and $E(\Omega) = -T(\Omega)/2$;
- $\lambda_1(\Omega)$, the first Dirichlet eigenvalue,

$$\lambda_1(\Omega) = \min_{0\neq v\in H_0^1(\Omega)} \frac{\int_\Omega |\nabla v|^2\,dx}{\int_\Omega v^2\,dx}, \tag{1}$$

  attained by a first eigenfunction $u > 0$ solving $-\Delta u = \lambda_1(\Omega)u$;
- B^*, the ball centred at 0 with $|B^*| = |\Omega|$; B_r the open ball of radius r centred at 0; B_1 the unit ball; $\omega_n = |B_1|$;
- $L_n = \lambda_1(B_1) = j_{n/2-1,1}^2$ and $T_n = T(B_1) = \omega_n/\big(n(n+2)\big)$;
- $\mathcal{A}(\Omega) = \inf_{x_0\in\mathbb{R}^n} |\Omega \,\triangle\, (x_0 + B^*)| \,/\, |B^*| \in [0, 2)$, the Fraenkel asymmetry;
- $\delta_{\mathrm{FK}}(\Omega) = (|\Omega|\,/\omega_n)^{2/n}\big(\lambda_1(\Omega) - \lambda_1(B^*)\big)$, the scale-invariant Faber–Krahn deficit.

The torsion function of a ball is $u_{B_r}(x) = (r^2 - |x|^2)/(2n)$, whence

$$T(B_r) = \frac{\omega_n}{n(n+2)}\,r^{\,n+2} = T_n\,r^{\,n+2}. \tag{2}$$

Under dilation $T(t\Omega) = t^{n+2}T(\Omega)$ and $\lambda_1(t\Omega) = t^{-2}\lambda_1(\Omega)$, so

$$\Psi(\Omega) := T(\Omega)\,\lambda_1(\Omega)^{(n+2)/2} \tag{3}$$

is scale invariant; in particular $\Psi(B_r) = \Psi(B_1) = T_n L_n^{(n+2)/2}$ for every ball. We abbreviate $\beta := (n+2)/2$.

1.2. Part 1: the Kohler–Jobin inequality, proved from scratch.

Theorem 1.1 (Kohler–Jobin)**.** *For every open $\Omega \subset \mathbb{R}^n$ of finite measure with $\lambda_1(\Omega) < \infty$ and every ball $B \subset \mathbb{R}^n$,*

$$T(\Omega)\,\lambda_1(\Omega)^\beta \;\geq\; T(B)\,\lambda_1(B)^\beta, \qquad \beta = \tfrac{n+2}{2}. \tag{4}$$

Equivalently $\Psi(\Omega) \geq \Psi(B_1) = T_n L_n^\beta$, with equality if and only if Ω is a ball (up to translation and a Lebesgue-null modification).

Strategy. Both sides of (4) have the same scaling and $\Psi(B) = \Psi(B_1)$ for every ball; so it suffices to compare Ω with a single, well-chosen ball. The idea is to rearrange the first eigenfunction u of Ω into a radial decreasing function u^* on a ball B in such a way that

(A) the Dirichlet energy is *exactly preserved*, $\int_B |\nabla u^*|^2 = \int_\Omega |\nabla u|^2$, and
(B) the L^2 norm *does not decrease*, $\int_B (u^*)^2 \geq \int_\Omega u^2$.

Granting (A)–(B), the Rayleigh principle (1) on B gives

$$\lambda_1(B) \;\le\; \frac{\int_B |\nabla u^*|^2}{\int_B (u^*)^2} \;\le\; \frac{\int_\Omega |\nabla u|^2}{\int_\Omega u^2} = \lambda_1(\Omega), \tag{5}$$

the last equality because u is the first eigenfunction of Ω.

The decisive design choice is which geometric quantity of the superlevel sets the ball B is built to match. Schwarz symmetrisation matches *volume* and yields the Faber–Krahn inequality directly; the Kohler–Jobin rearrangement instead matches a *torsional functional*, the *modified torsional rigidity* introduced next, and this is exactly what allows the Dirichlet energy to be preserved (not merely decreased) while the L^2 norm grows. The relation $\lambda_1(B) \le \lambda_1(\Omega)$ *at equal torsion* of B and Ω is then the inequality (4).

Standing regularity facts (cited, not proved).

(R1) *(Regularity of the eigenfunction.)* The first eigenfunction u of Ω may be taken positive, solves $-\Delta u = \lambda u$ in Ω with $\lambda = \lambda_1(\Omega)$, and satisfies $u \in C^1(\Omega) \cap L^\infty(\Omega)$ (Moser iteration for the L^∞ bound, De Giorgi–Nash for C^1; see [1, Prop. 2.4]). Put $M := \|u\|_{L^\infty(\Omega)} < \infty$.

(R2) *(Coarea formula.)* For $u \in W^{1,1}_{\mathrm{loc}}$ and Borel $g \ge 0$, $\int_\Omega g\, |\nabla u|\, dx = \int_0^M \big(\int_{\{u=t\}} g\, d\mathcal{H}^{n-1}\big)\, dt$ (Federer; see Maggi, *Sets of Finite Perimeter*, Thm. 13.1).

(R3) *(Saint–Venant inequality.)* For any open bounded $A \subset \mathbb{R}^n$, $T(A) \le T(A^\#)$ where $A^\#$ is the ball with $|A^\#| = |A|$; in the scale-free form we use it,

$$T(A) \;\le\; T_n\Big(\frac{|A|}{\omega_n}\Big)^{(n+2)/n}, \tag{6}$$

with equality iff A is a ball. ([1, (1.7) with $p = 2$].)

(R4) *(Distribution function.)* The distribution function $\mu_u(t) := |\{u > t\}|$ is non-increasing; together with the coarea formula this gives, for a.e. regular value t, that $\int_{\{u=t\}} |\nabla u|^{-1}\, d\mathcal{H}^{n-1}$ is finite and $\mu_u'(t) = -\int_{\{u=t\}} |\nabla u|^{-1}\, d\mathcal{H}^{n-1}$ (Brothers–Ziemer).

Two preliminary identities. We record the variational form of T and the level-set flux identity for the eigenfunction; both are elementary.

Lemma 1.2 (Variational torsion and the eigenfunction flux)**.**

(i) *For open A of finite measure, with $F(w) := \int_A w\, dx - \frac{1}{2}\int_A |\nabla w|^2\, dx$,*

$$T(A) = 2 \max_{w \in H^1_0(A)} F(w), \tag{7}$$

the maximiser being the torsion function u_A; equivalently $T(A) = \max\{(\int_A w)^2 / \int_A |\nabla w|^2 : 0 \ne w \in H^1_0(A)\}$.

(ii) *The first eigenfunction u satisfies, for a.e. $t \in (0, M)$,*

$$\int_{\{u=t\}} |\nabla u|\, d\mathcal{H}^{n-1} = \lambda \int_{\{u>t\}} u\, dx. \tag{8}$$

Proof. (i): F is strictly concave and coercive on $H^1_0(A)$; its Euler equation is $-\Delta w = 1$, so the maximiser is u_A and $F(u_A) = \int u_A - \frac{1}{2}\int |\nabla u_A|^2 = \int u_A - \frac{1}{2}\int u_A = \frac{1}{2}\int u_A = \frac{1}{2}T(A)$, using $\int |\nabla u_A|^2 = \int u_A$ (test the equation by u_A). The second form is the standard Rayleigh-type characterisation of T ([1, Prop. 2.2] with $p = 2$).

(ii): integrate $-\Delta u = \lambda u$ over the (a.e. smooth) superlevel set $\{u > t\}$ and use the divergence theorem with outward normal $\nu = -\nabla u / |\nabla u|$ on $\partial\{u > t\} = \{u = t\}$: $\int_{\{u=t\}} |\nabla u|\, d\mathcal{H}^{n-1} = -\int_{\{u=t\}} \partial_\nu u\, d\mathcal{H}^{n-1} = \int_{\{u>t\}} (-\Delta u)\, dx = \lambda \int_{\{u>t\}} u\, dx$. □

The modified torsional rigidity. The engine of the technique is the following functional, which assigns to a nonnegative function a torsion-type quantity that is sensitive to the *level structure* of the function, not just to its domain.

Definition 1.3 (Modified torsional rigidity)**.** Let $u \in H_0^1(\Omega) \cap L^\infty(\Omega)$, $u \geq 0$, $M = \|u\|_\infty$, with distribution function $\mu(t) = |\{u > t\}|$. Assume the integrability condition

$$t \longmapsto \frac{\mu(t)}{\int_{\{u=t\}} |\nabla u| \, d\mathcal{H}^{n-1}} \in L^\infty(0, M) \tag{9}$$

(satisfied by torsion functions and by first eigenfunctions, by (8); cf. [1, Lem. 3.2]). With $\mathcal{L} := \{g \in \mathrm{Lip}([0, M]) : g(0) = 0\}$, define the *modified torsional rigidity of* Ω *along* u

$$\mathrm{T}_{\mathrm{mod}}(\Omega; u) := 2 \sup_{g \in \mathcal{L}} F\big(g \circ u\big), \qquad F(w) = \int_\Omega w \, dx - \tfrac{1}{2} \int_\Omega |\nabla w|^2 \, dx. \tag{10}$$

Since the competitors $g \circ u$ form a subclass of $H_0^1(\Omega)$, comparison with (7) gives $\mathrm{T}_{\mathrm{mod}}(\Omega; u) \leq T(\Omega)$ always.

Lemma 1.4 (Closed form of $\mathrm{T}_{\mathrm{mod}}$)**.** *Under the hypotheses of Definition 1.3, writing* $P(t) := \int_{\{u=t\}} |\nabla u| \, d\mathcal{H}^{n-1}$*, the supremum in* (10) *is uniquely attained by the nondecreasing* g_0 *with* $g_0'(t) = \mu(t)/P(t)$*, and*

$$\mathrm{T}_{\mathrm{mod}}(\Omega; u) = \int_0^M \frac{\mu(t)^2}{P(t)} \, dt. \tag{11}$$

Proof. We may restrict to nondecreasing g: replacing g by $\tilde{g}(t) = \int_0^t |g'|$ leaves $\int_\Omega |\nabla(g \circ u)|^2$ unchanged (since $\tilde{g}' = |g'|$ a.e.) and increases $\int_\Omega g \circ u$ (since $\tilde{g} \geq g$), hence increases F. For nondecreasing $g \in \mathcal{L}$, Cavalieri's principle gives $\int_\Omega g(u) \, dx = \int_0^M g'(t)\mu(t) \, dt$, while the coarea formula (R2) gives $\int_\Omega |\nabla(g \circ u)|^2 \, dx = \int_0^M g'(t)^2 P(t) \, dt$. Thus

$$F(g \circ u) = \int_0^M \Big[g'(t)\mu(t) - \tfrac{1}{2} g'(t)^2 P(t)\Big] \, dt \leq \int_0^M \max_{s \geq 0} \Big[s\, \mu(t) - \tfrac{1}{2} s^2 P(t)\Big] \, dt = \int_0^M \frac{\mu(t)^2}{2P(t)} \, dt,$$

the pointwise maximum $\max_s(s\mu - \frac{1}{2}s^2 P) = \mu^2/(2P)$ attained at $s = \mu(t)/P(t)$. Equality holds iff $g'(t) = \mu(t)/P(t)$ a.e., which by (9) is an L^∞ function, so $g_0 \in \mathcal{L}$. Multiplying by 2 gives (11). $\square$

The next proposition is the isoperimetric statement that makes the rearrangement decrease the Rayleigh quotient. It is the only place where the Saint–Venant inequality enters.

Proposition 1.5 (Isoperimetric inequality for $\mathrm{T}_{\mathrm{mod}}$)**.** *Let* $u \in H_0^1(\Omega) \cap L^\infty$*,* $u \geq 0$*, satisfy* (9)*. If* $B \subset \mathbb{R}^n$ *is a ball with* $T(B) = \mathrm{T}_{\mathrm{mod}}(\Omega; u)$*, then* $|B| \leq |\Omega|$*, with equality iff* Ω *is a ball and* u *is (a translate of) a radial function.*

Proof. $T(B) = \mathrm{T}_{\mathrm{mod}}(\Omega; u) \leq T(\Omega)$ by the remark after Definition 1.3. The Saint–Venant inequality (6) gives $T(\Omega) \leq T_n(|\Omega| / \omega_n)^{(n+2)/n}$, while $T(B) = T_n(|B| / \omega_n)^{(n+2)/n}$ by (2). Hence $(|B| / \omega_n)^{(n+2)/n} \leq (|\Omega| / \omega_n)^{(n+2)/n}$, i.e. $|B| \leq |\Omega|$. Equality forces $\mathrm{T}_{\mathrm{mod}}(\Omega; u) = T(\Omega)$ and equality in Saint–Venant, hence Ω a ball; the maximiser $g_0 \circ u$ of (10) then coincides with the (radial) torsion function, so u is radial. ([1, Prop. 3.8].) $\square$

The Kohler–Jobin rearrangement. We now build u^*. For $t \in [0, M]$ set $\Omega_t = \{u > t\}$ and let

$$\mathcal{T}(t) := \mathrm{T}_{\mathrm{mod}}\big(\Omega_t; (u - t)_+\big). \tag{12}$$

Since $|\{(u - t)_+ > s\}| = \mu(t + s)$ and the coarea factor of $(u - t)_+$ at level s is $P(t + s)$, formula (11) gives the explicit expression

$$\mathcal{T}(t) = \int_t^M \frac{\mu(\tau)^2}{P(\tau)} \, d\tau, \tag{13}$$

so that $\mathcal{T} \in \mathrm{Lip}_{\mathrm{loc}}([0, M))$, $\mathcal{T}(M) = 0$, $\mathcal{T}(0) = \mathrm{T}_{\mathrm{mod}}(\Omega; u) =: T_\star$, and for a.e. t

$$\mathcal{T}'(t) = -\frac{\mu(t)^2}{P(t)} < 0. \tag{14}$$

Thus $\mathcal{T}$ is a strictly decreasing bijection $[0, M] \to [0, T_\star]$.

Definition of u^*. Let $B = B_{R_\star}$ be the ball with $T(B) = T_\star$, i.e. $R_\star = (T_\star/T_n)^{1/(n+2)}$. For $\tau \in [0, T_\star]$ let $R(\tau)$ be the radius with $T(B_{R(\tau)}) = \tau$, i.e.

$$R(\tau) = (\tau/T_n)^{1/(n+2)}, \qquad \left|B_{R(\tau)}\right| = \omega_n R(\tau)^n = \omega_n (\tau/T_n)^{n/(n+2)}. \tag{15}$$

Let $\varphi := \mathcal{T}^{-1} : [0, T_\star] \to [0, M]$ (so $\varphi(0) = M$, $\varphi(T_\star) = 0$, $\varphi' < 0$). We define u^* on B to be the radial decreasing function whose value on the sphere $\{|x| = R(\tau)\}$ is $\psi(\tau)$, where $\psi : [0, T_\star] \to [0, \infty)$ is the Lipschitz function determined by

$$\bigl(-\psi'(\tau)\bigr)\, \mu^*\bigl(\psi(\tau)\bigr) = \bigl(-\varphi'(\tau)\bigr)\, \mu\bigl(\varphi(\tau)\bigr), \qquad \psi(T_\star) = 0, \tag{16}$$

μ^* denoting the distribution function of u^*. Concretely the superlevel set $\{u^* > \psi(\tau)\} = B_{R(\tau)}$ has, by construction, torsional rigidity exactly τ, the value of the *modified* torsion that the corresponding superlevel set $\Omega_{\varphi(\tau)}$ of u carries. In particular $\{u^* > 0\} = B = B_{R_\star}$, so

$$T(B) = T_\star = \mathrm{T}_{\mathrm{mod}}(\Omega; u) \quad \le \quad T(\Omega). \tag{17}$$

Since $\mu^*(\psi(\tau)) = \left|B_{R(\tau)}\right| = \omega_n R(\tau)^n$, equation (16) integrates to an explicit ψ; we shall only need the two qualitative consequences (A), (B), proved next.

Lemma 1.6 (Dirichlet energy is preserved). $\displaystyle\int_B |\nabla u^*|^2\, dx = \int_\Omega |\nabla u|^2\, dx.$

Proof. The original side. By coarea and the change of variable $t = \varphi(\tau)$ (Lemma 2.5 of [1], the one-dimensional area formula for $\varphi = \mathcal{T}^{-1} \in \mathrm{Lip}_{\mathrm{loc}}$ with $\varphi' < 0$),

$$\int_\Omega |\nabla u|^2\, dx = \int_0^M P(t)\, dt = \int_0^{T_\star} P(\varphi(\tau))\, (-\varphi'(\tau))\, d\tau. \tag{18}$$

Differentiating $\mathcal{T}(\varphi(\tau)) = \tau$ and using (14) gives the key relation

$$-\varphi'(\tau) = \frac{1}{-\mathcal{T}'(\varphi(\tau))} = \frac{P(\varphi(\tau))}{\mu(\varphi(\tau))^2}. \tag{19}$$

Hence, writing $P(\varphi)(-\varphi') = \mu(\varphi)^2 \cdot \frac{P(\varphi)}{\mu(\varphi)^2} \cdot (-\varphi')$ and using (19) for the middle factor, $P(\varphi)(-\varphi') = \mu(\varphi)^2(-\varphi')^2$, so (18) becomes

$$\int_\Omega |\nabla u|^2\, dx = \int_0^{T_\star} \mu(\varphi(\tau))^2\, (-\varphi'(\tau))^2\, d\tau. \tag{20}$$

The rearranged side. For the radial u^*, the level set $\{u^* = \psi(\tau)\}$ is the sphere of radius $R(\tau)$, on which $|\nabla u^*|$ is constant. A direct computation (differentiate $u^*(x) = \psi(\tau)$ on $|x| = R(\tau)$ and use $R(\tau) = (\tau/T_n)^{1/(n+2)}$) gives, with $P^*(\tau') := \int_{\{u^* = \tau'\}} |\nabla u^*|\, d\mathcal{H}^{n-1}$ the coarea factor of u^*,

$$\int_B |\nabla u^*|^2\, dx = \int_0^{T_\star} \mu^*(\psi(\tau))^2\, (-\psi'(\tau))^2\, d\tau, \tag{21}$$

the exact radial analogue of (20); this is the specialisation to $p = 2$ of the identity (4.4) of [1], obtained there by the same coarea computation. Now the defining relation (16), $\mu^*(\psi)(-\psi') = \mu(\varphi)(-\varphi')$, makes the integrands of (21) and (20) *equal*:

$$\mu^*(\psi(\tau))^2(-\psi'(\tau))^2 = \bigl[\mu^*(\psi(\tau))(-\psi'(\tau))\bigr]^2 = \bigl[\mu(\varphi(\tau))(-\varphi'(\tau))\bigr]^2 = \mu(\varphi(\tau))^2(-\varphi'(\tau))^2.$$

Hence $\int_B |\nabla u^*|^2 = \int_\Omega |\nabla u|^2$. □

Lemma 1.7 (L^2 norm does not decrease). $\displaystyle\int_B (u^*)^2\, dx \quad \ge \quad \int_\Omega u^2\, dx$, *and more generally* $\int_B f(u^*)\, dx \ge \int_\Omega f(u)\, dx$ *for every convex* $f : [0, \infty) \to [0, \infty)$ *with* $f(0) = 0$, *strictly if* f *is strictly convex unless* Ω *is a ball and* u *radial.*

Proof. The point is the comparison of *distribution functions* along the rearrangement. Apply Proposition 1.5 not to u but to the truncations: for each $\tau \in [0, T_\star]$ the ball $B_{R(\tau)}$ has $T(B_{R(\tau)}) = \tau = \mathrm{T}_{\mathrm{mod}}(\Omega_{\varphi(\tau)}; (u - \varphi(\tau))_+)$ by construction, so Proposition 1.5 (applied to $\Omega_{\varphi(\tau)}$ with reference function $(u - \varphi(\tau))_+$) yields

$$\mu^*(\psi(\tau)) = \left|B_{R(\tau)}\right| \;\le\; \left|\Omega_{\varphi(\tau)}\right| = \mu(\varphi(\tau)), \qquad \tau \in [0, T_\star]. \tag{22}$$

Combining (22) with the defining relation (16), $\mu^*(\psi)(-\psi') = \mu(\varphi)(-\varphi')$, gives

$$-\psi'(\tau) = \frac{\mu(\varphi(\tau))}{\mu^*(\psi(\tau))}\,(-\varphi'(\tau)) \;\ge\; -\varphi'(\tau) \qquad \text{for a.e. } \tau, \tag{23}$$

since $\mu(\varphi) \ge \mu^*(\psi)$. Integrating from τ to $T_\star$ and using $\psi(T_\star) = \varphi(T_\star) = 0$,

$$\psi(\tau) = \int_\tau^{T_\star} (-\psi'(s))\,ds \;\ge\; \int_\tau^{T_\star} (-\varphi'(s))\,ds = \varphi(\tau), \qquad \tau \in [0, T_\star]. \tag{24}$$

Now compute, for convex f with $f(0) = 0$ (so $f(u) = \int_0^u f'(t)\,dt$ and Cavalieri applies), using the change of variable $t = \varphi(\tau)$:

$$\begin{aligned}
\int_\Omega f(u)\,dx &= \int_0^M f'(t)\,\mu(t)\,dt = \int_0^{T_\star} f'(\varphi(\tau))\,(-\varphi'(\tau))\,\mu(\varphi(\tau))\,d\tau \\
&= \int_0^{T_\star} f'(\varphi(\tau))\,(-\psi'(\tau))\,\mu^*(\psi(\tau))\,d\tau \qquad \text{by (16)} \\
&\le \int_0^{T_\star} f'(\psi(\tau))\,(-\psi'(\tau))\,\mu^*(\psi(\tau))\,d\tau \qquad \text{by (24) and } f' \text{ nondecreasing} \\
&= \int_0^M f'(s)\,\mu^*(s)\,ds = \int_B f(u^*)\,dx,
\end{aligned}$$

the last change of variable being $s = \psi(\tau)$. Taking $f(s) = s^2$ proves $\int_B (u^*)^2 \ge \int_\Omega u^2$. If f is strictly convex, equality in the middle step forces $\psi = \varphi$ a.e., hence equality in (22), hence Ω a ball and u radial by the equality case of Proposition 1.5. $\square$

Proof of Theorem 1.1. Apply the construction to the first eigenfunction u of Ω (a valid reference function by (8) and (R1)). By Lemmas 1.6 and 1.7 the rearranged $u^* \in H_0^1(B)$ satisfies $\int_B |\nabla u^*|^2 = \int_\Omega |\nabla u|^2$ and $\int_B (u^*)^2 \ge \int_\Omega u^2$. By the Rayleigh principle on B (1) and these two facts,

$$\lambda_1(B) \le \frac{\int_B |\nabla u^*|^2}{\int_B (u^*)^2} \le \frac{\int_\Omega |\nabla u|^2}{\int_\Omega u^2} = \lambda_1(\Omega).$$

Multiplying $\lambda_1(B) \le \lambda_1(\Omega)$ by $T(B)^\beta > 0$ and using $T(B) \le T(\Omega)$ from (17) and the scale invariance $\Psi(B) = \Psi(B_1)$:

$$T(B_1)L_n^\beta = \Psi(B) = T(B)\lambda_1(B)^\beta \le T(\Omega)\lambda_1(\Omega)^\beta = \Psi(\Omega).$$

This is (4) (for an arbitrary ball, $\Psi(B) = \Psi(B_1)$ on both sides). Equality forces equality throughout, in particular in Lemma 1.7 with the strictly convex $f(s) = s^2$, hence Ω is a ball. $\square$

Remark 1.8 (On the sign that distinguishes Kohler–Jobin from Faber–Krahn)**.** The torsion-preserving rearrangement makes the ball $B_{R(\tau)}$ *smaller* than the superlevel set $\Omega_{\varphi(\tau)}$, (22) — the opposite of Schwarz symmetrisation. It is precisely this that lets the Dirichlet energy be *exactly preserved* (Lemma 1.6) rather than merely decreased, while the level-set volumes *grow under the inverse rearrangement*, forcing $\int (u^*)^2 \ge \int u^2$ (Lemma 1.7). The naive attempt to compare L^2 masses level by level fails because the rearranged superlevel balls are smaller; the correct comparison is through the *distribution-function inequality* (24), driven by the isoperimetric Proposition 1.5.

1.3. **Part 2: transfer to sharp Faber–Krahn stability.** We now use Theorem 1.1 as a tool. Throughout, $L := \lambda_1(B^*)$, $T_0 := T(B^*)$, $\beta = (n+2)/2$, $p := 1/\beta = 2/(n+2) \in (0,1]$.

Lemma 1.9 (Pointwise Kohler–Jobin transfer)**.** *For every open $\Omega \subset \mathbb{R}^n$ with $|\Omega| = |B^*|$,*

$$\lambda_1(\Omega) \;\geq\; L\Big(\frac{T_0}{T(\Omega)}\Big)^{1/\beta} = L\Big(\frac{T_0}{T(\Omega)}\Big)^{2/(n+2)}. \tag{25}$$

Proof. Theorem 1.1 with $B = B^*$ gives $T(\Omega)\lambda_1(\Omega)^\beta \geq T_0 L^\beta$; solve for $\lambda_1(\Omega)$. □

By Saint–Venant (6) at fixed volume, $T(\Omega) \leq T_0$, so the torsion deficit $\sigma := \sigma(\Omega) := T_0 - T(\Omega) \geq 0$. Put $s := \sigma/T_0 \in [0,1)$; then (25) reads

$$\lambda_1(\Omega) - L \;\geq\; L\big[(1-s)^{-p} - 1\big], \qquad p = \tfrac{2}{n+2}. \tag{26}$$

Lemma 1.10 (An elementary one-variable inequality)**.** *For $p \in (0,1]$ and $s \in [0,1)$, $(1-s)^{-p} - 1 \geq p\,s$.*

Proof. $f(s) = (1-s)^{-p} - 1$ has $f(0) = 0$ and $f'(s) = p(1-s)^{-(p+1)} \geq p$ on $[0,1)$ because $(1-s)^{-(p+1)} \geq 1$. By the fundamental theorem of calculus $f(s) = \int_0^s f' \geq ps$. □

Lemma 1.11 (Small-deficit linearisation)**.** *If $|\Omega| = |B^*|$ and $\sigma \leq T_0/2$, then $\lambda_1(\Omega) - L \geq \dfrac{2L}{(n+2)T_0}\sigma$.*

Proof. With $s = \sigma/T_0 \leq 1/2$, $p = 2/(n+2)$, combine (26) and Lemma 1.10: $\lambda_1(\Omega) - L \geq Lps = Lp\sigma/T_0 = \frac{2L}{(n+2)T_0}\sigma$. □

Lemma 1.12 (Large-deficit branch)**.** *If $|\Omega| = |B^*|$ and $\sigma > T_0/2$, then $\lambda_1(\Omega) - L \geq L(2^{2/(n+2)} - 1)$, hence, since $\mathcal{A}(\Omega) < 2$,*

$$\lambda_1(\Omega) - L \;\geq\; \frac{L(2^{2/(n+2)} - 1)}{4}\,\mathcal{A}(\Omega)^2. \tag{27}$$

Proof. $\sigma > T_0/2$ gives $T(\Omega) < T_0/2$, so $T_0/T(\Omega) > 2$ and (25) yields $\lambda_1(\Omega) \geq L\,2^{2/(n+2)}$. Since $\mathcal{A}(\Omega)^2 \leq 4$ at fixed volume, dividing by 4 gives (27). □

Theorem 1.13 (Sharp Faber–Krahn via Kohler–Jobin)**.** *Assume the global sharp Saint–Venant stability estimate: there is $c_{\mathrm{SV}}^{\mathrm{glob}}(n) > 0$ such that*

$$T(B^*) - T(\Omega) = 2\big(E(\Omega) - E(B^*)\big) \;\geq\; 2\,c_{\mathrm{SV}}^{\mathrm{glob}}(n)\Big(\frac{|\Omega|}{\omega_n}\Big)^{(n+2)/n}\mathcal{A}(\Omega)^2 \tag{28}$$

for all open bounded $\Omega \subset \mathbb{R}^n$, $0 < |\Omega| < \infty$. Then

$$\delta_{\mathrm{FK}}(\Omega) = \Big(\frac{|\Omega|}{\omega_n}\Big)^{2/n}\big(\lambda_1(\Omega) - \lambda_1(B^*)\big) \;\geq\; c_{\mathrm{FK}}(n)\,\mathcal{A}(\Omega)^2, \tag{29}$$

with the explicit constant

$$c_{\mathrm{FK}}(n) = \min\left\{\frac{4\,L_n\,c_{\mathrm{SV}}^{\mathrm{glob}}(n)}{(n+2)\,T_n},\;\; \frac{L_n\big(2^{2/(n+2)} - 1\big)}{4}\right\}, \tag{30}$$

equivalently $\mathcal{A}(\Omega)^2 \leq C_{\mathrm{FK}}(n)\,\delta_{\mathrm{FK}}(\Omega)$ with $C_{\mathrm{FK}}(n) = 1/c_{\mathrm{FK}}(n)$.

Proof. Both δ_{FK} and $\mathcal{A}$ are scale invariant, and so is (28); rescale so that $|\Omega| = \omega_n$, whence $B^* = B_1$, $L = L_n$, $T_0 = T_n$, and the volume prefactor $(|\Omega|/\omega_n)^{2/n} = 1$. Split on the torsion deficit $\sigma = T_n - T(\Omega)$.

Case 1 ($\sigma \leq T_n/2$). By Lemma 1.11 and (28) (which now reads $\sigma \geq 2c_{\mathrm{SV}}^{\mathrm{glob}}(n)\mathcal{A}(\Omega)^2$),

$$\lambda_1(\Omega) - L_n \geq \frac{2L_n}{(n+2)T_n}\sigma \geq \frac{2L_n}{(n+2)T_n}\cdot 2c_{\mathrm{SV}}^{\mathrm{glob}}(n)\,\mathcal{A}(\Omega)^2 = \frac{4L_n c_{\mathrm{SV}}^{\mathrm{glob}}(n)}{(n+2)T_n}\,\mathcal{A}(\Omega)^2.$$

Case 2 ($\sigma > T_n/2$). By Lemma 1.12, $\lambda_1(\Omega) - L_n \geq \frac{L_n(2^{2/(n+2)}-1)}{4}\mathcal{A}(\Omega)^2$.

In both cases $\delta_{\mathrm{FK}}(\Omega) = \lambda_1(\Omega) - L_n \geq c_{\mathrm{FK}}(n)\mathcal{A}(\Omega)^2$ with $c_{\mathrm{FK}}(n)$ the minimum (30). Undoing the scaling preserves the inequality since every term is scale invariant. □

Remark 1.14 (Sharpness of the exponent)**.** The exponent 2 on $\mathcal{A}$ in (29) is optimal: smooth ellipsoidal perturbations of the ball have $\lambda_1(\Omega) - L_n \asymp \mathcal{A}(\Omega)^2$ ([2, 1] and references therein). The transfer is non-perturbative: no compactness, contradiction, or Taylor expansion is used past the elementary Lemma 1.10.

References.

References

[1] L. Brasco, *On torsional rigidity and principal frequencies: an invitation to the Kohler–Jobin rearrangement technique*, ESAIM Control Optim. Calc. Var. **20** (2014), 315–338.

[2] L. Brasco, G. De Philippis, *Spectral inequalities in quantitative form*, in *Shape Optimization and Spectral Theory*, De Gruyter, 2017, pp. 201–281.

[3] M.-T. Kohler-Jobin, *Une méthode de comparaison isopérimétrique de fonctionnelles de domaines de la physique mathématique*, Z. Angew. Math. Phys. **29** (1978), 757–766.

[4] M.-T. Kohler-Jobin, *Démonstration de l'inégalité isopérimétrique $P\lambda^2 \geq \pi j_0^4/2$ conjecturée par Pólya et Szegő*, C. R. Acad. Sci. Paris **281** (1975), 119–121.

[5] M.-T. Kohler-Jobin, *Symmetrization with equal Dirichlet integrals*, SIAM J. Math. Anal. **13** (1982), 153–161.